

02425 Diffusions and Stochastic Differential Equations

Exercises Week 2: Probability Measures

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Consider the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with $\Omega = [0, 1]^2$, $\mathcal{F}$ the usual Borel-algebra on Ω , and $\mathbb{P}$ the Lebesgue measure, i.e. area. For $\omega = (x, y) \in \Omega$, define $X(\omega) = x$, $Y(\omega) = y$, and $Z(\omega) = x + y$.

Question 1, independence and marginal distribution of X and Y : Verify that X and Y are independent, and that each are uniformly distributed on $[0, 1]$;

We take the event $\{X \leq a\}$ which means $x \leq a$. But y is not restricted so $0 \leq y \leq 1$. The area of X is then given by $\mathbb{P}(X \leq a) = 1 \cdot a = a$. This proves X is uniformly distributed on $[0, 1]$ since $\mathbb{P}(X \leq a) = a$ for a between 0 and 1. The exact same proof is valid for Y .

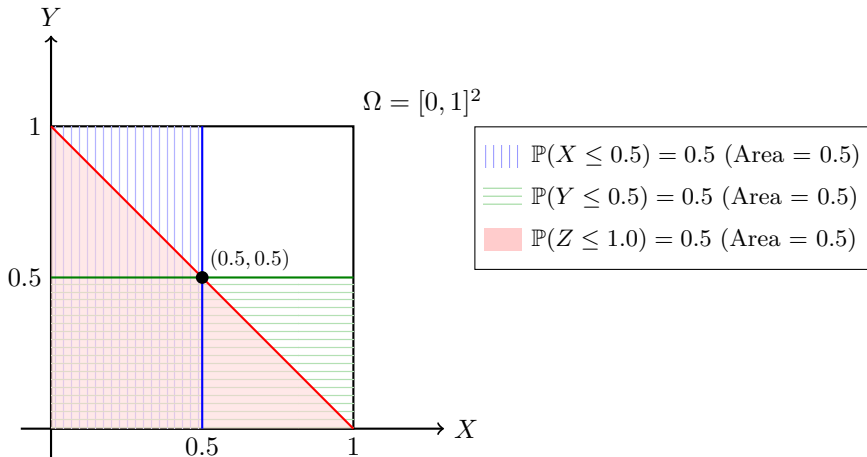
For the event $(X \leq a, Y \leq b)$ we again evaluate $\mathbb{P}$ as $\mathbb{P}(X \leq a, Y \leq b) = ab$. Now the probabilities of X and Y are $\mathbb{P}(X \leq a)$ and $\mathbb{P}(Y \leq b) = b$ as shown above. So by the definition of independence;

$$\mathbb{P}(X \leq a, Y \leq b) = \mathbb{P}(X \leq a)\mathbb{P}(Y \leq b)$$

Thus, X and Y are independent.

Question 2, level sets and σ -algebras: Sketch level sets (contour lines) for X , Y , and Z . *Note:* A level set for a function is a subset of the domain where the function attains one specific value, such as $X^{-1}(\frac{1}{2}) = \{(x, y) : x = \frac{1}{2}\}$. Show typical elements in the σ -algebras $\sigma(X)$, $\sigma(Y)$, and $\sigma(Z)$;

The sketches of $X^{-1}(\frac{1}{2})$, $Y^{-1}(\frac{1}{2})$, and $Z^{-1}(1)$;



To show typical elements of the σ -algebras we show the collection of all events for a random variable $\sigma(X)$ as $X^{-1}(B)$ where B is a Borel set. Since Borel sets are built from basic intervals, a typical element of the σ -algebra is then the preimage of a standard interval $[a, b]$.

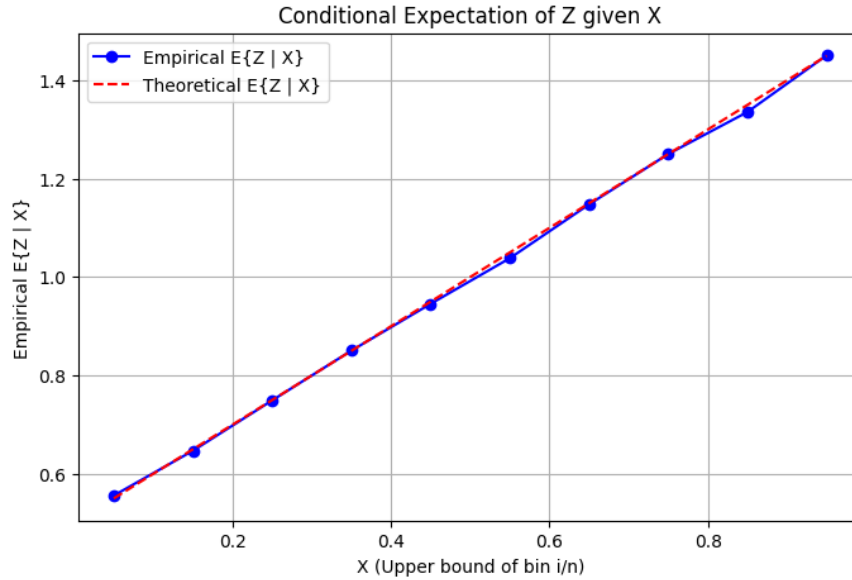
- For $\sigma(X)$ a typical element is $X^{-1}([a, b]) = \{(x, y) \in [0, 1]^2 \mid a \leq x \leq b\}$. This is all vertical level sets between a and b , and forms a vertical rectangle on the board.
- For $\sigma(Y)$ a typical element is $Y^{-1}([a, b]) = \{(x, y) \in [0, 1]^2 \mid a \leq y \leq b\}$. This is all horizontal level sets between a and b , and forms a horizontal rectangle on the board.
- For $\sigma(Z)$ a typical element is $Z^{-1}([a, b]) = \{(x, y) \in [0, 1]^2 \mid a \leq x + y \leq b\}$. This is all diagonal level sets between $x + y = a$ and $x + y = b$, and forms a diagonal band cutting across the sample space.

Question 3, Monte Carlo simulation and conditional expectations of Z given X Simulate a larger number, for example $N = 10,000$, of realizations of X , Y , and Z . Binning the realized values of X into bins $[0, \frac{1}{n}, \frac{2}{n}, \dots, 1]$, estimate empirically the conditional expectations of Z given X

$$\mathbb{E} \left\{ Z \mid X \in \left[\frac{i-1}{n}, \frac{i}{n} \right] \right\}$$

for $n = 10$ and $i = 1, \dots, n$. Plot the result against $\frac{1}{n}$;

After generating 10,000 x and y values, and calculating their corresponding z values, we mask out entries in bins between $\frac{i-1}{n}$ and $\frac{i}{n}$ for $n = 10$. Plotting Z given X against the theoretical upper bound $\frac{1}{n}$;

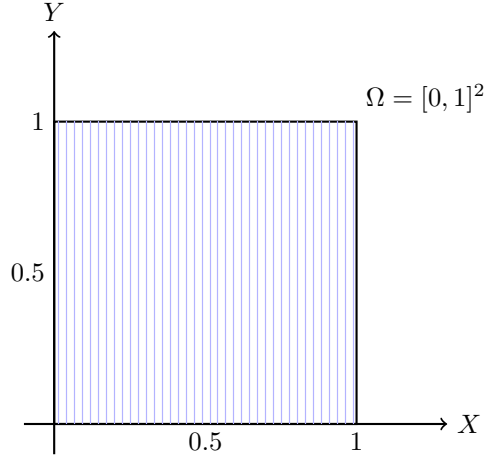


Question 4, conditional expectations of Z given X , analytically Find a function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that $\mathbb{E}\{Z|X\} = g(X)$. Use inspiration (!) from the Monte Carlo experiments and elementary arguments. Sketch contour lines of $\mathbb{E}\{Z|X\}$. Verify that this "candidate" conditional expectation satisfies the defining property

$$\int_H g(X(\omega)) d\mathbb{P}(\omega) = \int_H Z(\omega) d\mathbb{P}(\omega)$$

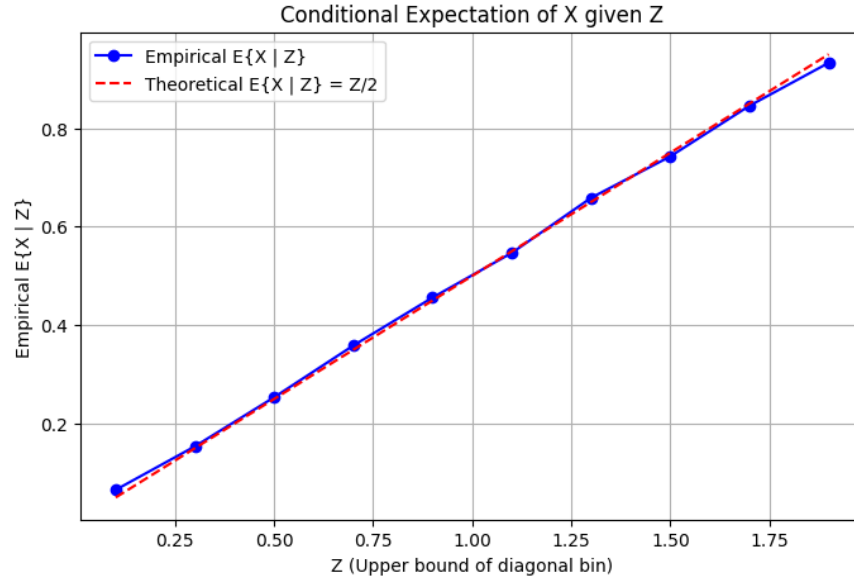
for a typical $H \in \sigma(X)$;

We know X and Y are independent. So if X is given (as $X = x$), then Y must be uniformly distributed for that given X . This gives that $\mathbb{E}\{Y|X\} = 0.5$, such that $\mathbb{E}\{Z|X\} = X + 0.5$. Thus, $g(x) = x + \frac{1}{2}$.



The level sets drawn in contour lines of $\mathbb{E}\{Z|X\}$ are the same as for X but with an offset of 0.5. Thus, $g(X)$ is $\sigma(X)$ -measurable; it perfectly respects the vertical information structure of X .

Question 5, conditional expectations of X given Z Repeat the two previous questions for $\mathbb{E}\{X|Z\}$: First estimate the function $h : \mathbb{R} \rightarrow \mathbb{R}$ such that $h(Z) = \mathbb{E}\{X|Z\}$ from the Monte Carlo experiments. Then find that function and verify that it satisfies its defining properties;



This shows the tendency that $\mathbb{E}\{X|Z\} = \frac{Z}{2}$. We can confirm this from the fact that X and Y are independent and identically distributed (i.i.d.), and thus $\mathbb{E}\{Z|Z\} = \mathbb{E}\{X|Z\} + \mathbb{E}\{Y|Z\}$ for which $\mathbb{E}\{X|Z\} = \mathbb{E}\{Y|Z\}$. So $h(Z) = \frac{Z}{2}$.

We must now prove

$$\int_H \frac{Z}{2} d\mathbb{P} = \int_H X d\mathbb{P}$$

And since we have symmetry between X and Y , we can establish that also $\int_H X d\mathbb{P} = \int_H Y d\mathbb{P}$. So if we

substitute $Z = X + Y$ we get;

$$\begin{aligned}\int_H \frac{X+Y}{2} d\mathbb{P} &= \int_H X d\mathbb{P} \\ \frac{1}{2} \int_H X d\mathbb{P} + \frac{1}{2} \int_H Y d\mathbb{P} &= \int_H X d\mathbb{P} \\ \frac{1}{2} \int_H X d\mathbb{P} + \frac{1}{2} \int_H X d\mathbb{P} &= \int_H X d\mathbb{P} \\ \int_H X d\mathbb{P} &= \int_H X d\mathbb{P}\end{aligned}$$

Thus, proving the property.

Question 6, the law of total expectation (the simple tower property) Verify from the Monte Carlo experiments that

$$\mathbb{E}\{\mathbb{E}\{X|Z\}\} = \mathbb{E}\{X\}$$

To verify the simple tower property from our Monte Carlo experiments, we evaluate both sides of the equation. The right side $\mathbb{E}\{X\}$ is calculated as the mean of the Python array `X = np.random.uniform(0, 1, 10_000)`. This is evaluated as the numerical value 0.4982, which is very close to the expected theoretical 0.5. For the left side $\mathbb{E}\{\mathbb{E}\{X|Z\}\}$ we remember that the inner term $\mathbb{E}\{X|Z\} = \frac{Z}{2}$. This means we evaluate the numerical value as the mean of $\frac{Z}{2}$, which gives us 0.4996. Since both calculations converge to 0.5 proves the simple tower property; averaging the conditional average gives us back the global average.

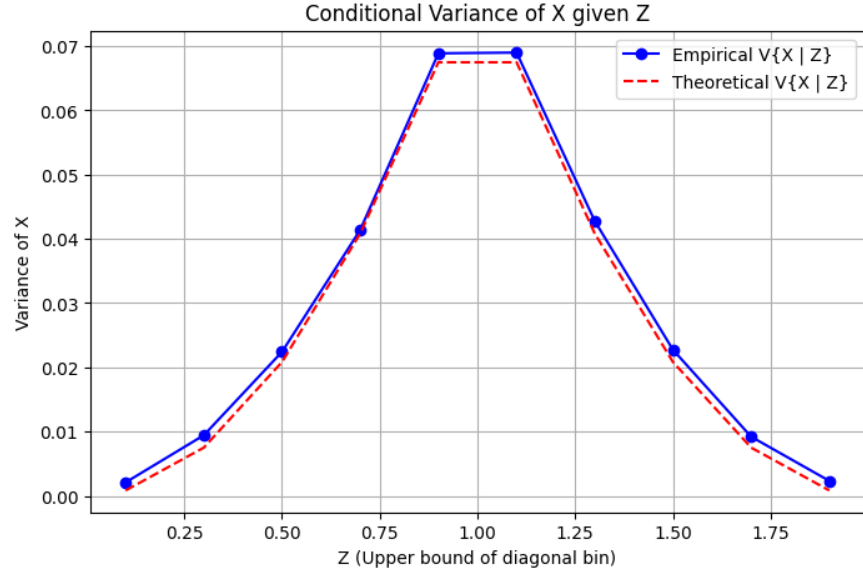
Question 7, variance decomposition/law of total variance Find an analytical expression for $\mathbb{V}\{X|Z\}$ and verify it against the Monte Carlo simulations. Then verify from the Monte Carlo simulations the variance decomposition formula

$$\mathbb{V}\{X\} = \mathbb{E}\mathbb{V}\{X|Z\} + \mathbb{V}\mathbb{E}\{X|Z\}$$

If we divide $\mathbb{V}\{X|Z\}$ into two cases; $z \leq 1$ and $z > 1$. Then we know the variance of a continuous uniform distribution to be the total length of the interval. This means that for $z \leq 1$ we establish that $0 \leq X \leq Z$, resulting in the length $Z - 0 = Z$. Likewise, for the second case we get that $Z - 1 \leq X \leq 1$, such that $1 - (Z - 1) = 2 - Z$. So using the formula $\mathbb{V} = \frac{(b-a)^2}{12}$ we get;

$$\mathbb{V}\{X|Z\} = \begin{cases} \frac{z^2}{12} & \text{if } z \leq 1 \\ \frac{(2-z)^2}{12} & \text{if } z > 1 \end{cases}$$

This means the variance is 0 at the extremes $Z = 0$ and $Z = 2$, where only one exact coordinate results in that sum. And the variance is highest in the middle where $Z = 1$, where the diagonal of level sets is the longest, and X has the most room to vary. Simulating the variance yields following plot;



Verifying the Monte Carlo simulations for the variance decomposition formula we get that $V\{X\} = 0.0827$, $\mathbb{E}V\{X|Z\} = 0.0419$, and $V\mathbb{E}\{X|Z\} = 0.0409$. This gives us $0.0419 + 0.0409 \approx 0.0827$, which is true with a little allowed noise from the Monte Carlo simulations.